CHICAGO VEHICLE CRASHES

Building a model to predict the primary contributory cause of car accidents particularly in Chicago

Step 1: Business Understanding

The goal of this project is to analyze vehicle crash data to **identify the key factors that contribute to road accidents** which will help public safety agencies and policy makers enhance road safety and recommend actionable insights to reduce crashes, injuries, and fatalities.

Overview/Background

Road traffic accidents are a major public safety issue in urban environments, leading to thousands of injuries, fatalities, and significant economic loss each year. In cities like Chicago, understanding the factors that contribute to vehicle crashes is critical for designing effective transportation policies, improving road safety and reducing preventable deaths.

This project leverages historical crash data to:

- Explore trends in crash severity;
- Identify key contributory factors such as driver age, vehicle condition, or environmental conditions;
- Provide insights that support targeted interventions such as safer road designs, better driver education, improved enforcement strategies, and smarter resource allocation.

Problem Statement

Road traffic crashes remain a significant public safety issue often resulting in severe injuries or fatalities. To effectively reduce crash occurrences and improve safety outcomes, it is essential to understand the key factors that contribute to crash severity and frequency. This analysis aims to explore this large-scale traffic crash dataset to answer critical questions. By addressing these questions, the analysis will help identify high-risk conditions, behaviors and regions, enabling data-driven interventions and policy recommendations aimed at improving road safety and reducing crash-related injuries and fatalities.

Objectives

1. Identify and analyse key risk factors contributing to crash severity.
2. Examine temporal and situational patterns in crash and injury severity.
3. Evaluate the impact of preventive measures on crash outcomes.

Step 2: Data Understanding

The following datasets were provided;

- cpd-traffic-crashes Crash data shows information about each traffic crash on city streets within the City of Chicago limits and under the jurisdiction of Chicago Police Department (CPD).
- cpd-traffic-crashes-people This data contains information about people involved in a crash and if any injuries were sustained.
- cpd-traffic-crashes-vehicles This dataset contains information about vehicles (or units as they are identified in crash reports) involved in a traffic crash.

The three datasets were utilized to gain a comprehensive understanding of crash data involving both individuals and vehicles. The datasets were merged into a single DataFrame containing records and features, forming the basis for subsequent data preprocessing and cleaning.

Step 3: Data preparation

- To ensure high-quality analysis and reliable model results, the data from three distinct sources was carefully **loaded, merged, cleaned, and filtered**.
- Using CRASH_RECORD_ID, the three datasets were merged into a single comprehensive DataFrame that represents **crash events**, with enriched information about **drivers/passengers and vehicles**.
- A missing value analysis was performed to identify and quantify nulls. Visual tools like **heatmaps** and **bar plots** were used to detect patterns in missingness.
- Data cleaning to ensure unrealistic entries were removed, columns with missing values were dropped, categorical columns were retained for encoding.
- A cleaned, integrated and analysis dataset was produced – df_clean.

Step 4: Modeling and Evaluation

The objective of the being to determine the primary contributory cause using three different datasets means that the problem is formulated as a **multi-class classification problem**, where each crash is assigned one of several possible contributory causes.

- We did split the dataset into training set and testing set.
- We used the following models in our prediction Logistic Regression, Decision Tree, Random Forest, Ada Boost, XG Boost. Build pipeline for each model.
- Performed Grid Search to optimize hyper parameters
- Since it was a multi-class classification we used the following metrics to evaluate our models; accuracy, precision, recall, F1-Score.

Step 5: Deployment

After training and evaluating the model to predict vehicle crash contributory causes or injury severity, the next step is to **deploy** the solution so it can be used by stakeholders like transportation departments, safety analysts, or emergency responders.

Important Links

Github: https://github.com/billysambasi/Group7_Phase4_Project